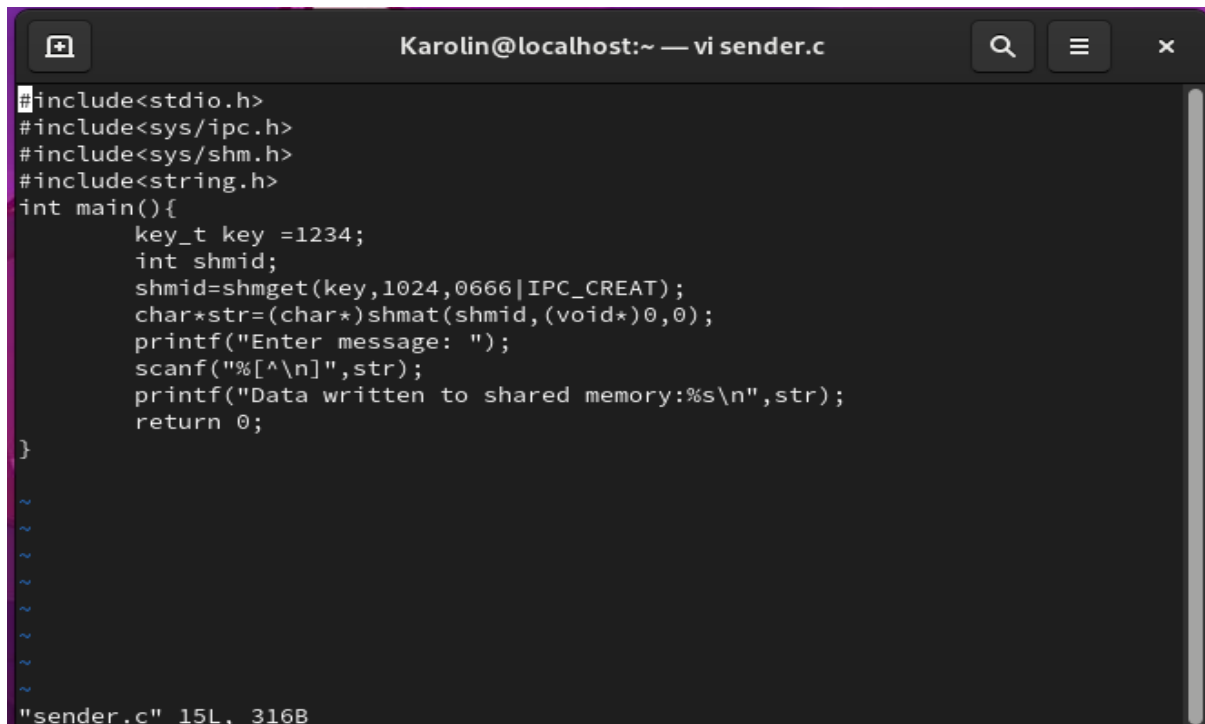
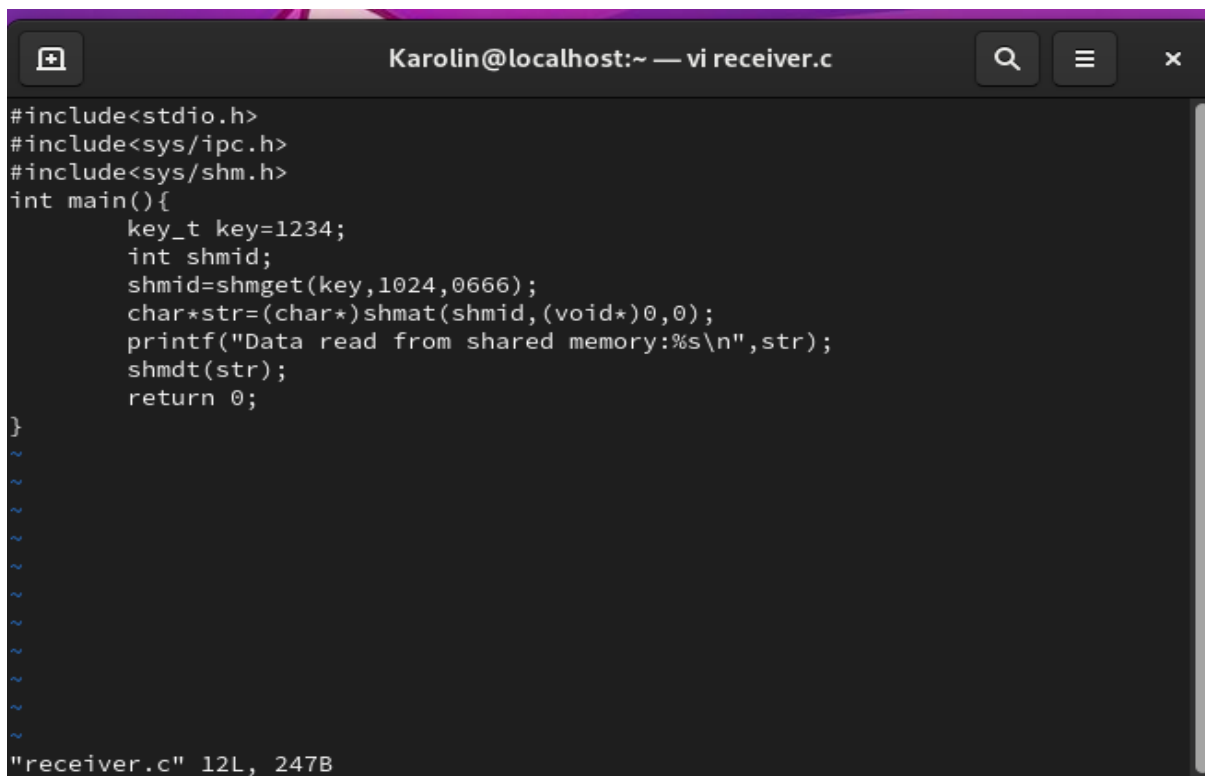


Ex. 7 Inter Process Communication Using Semaphores



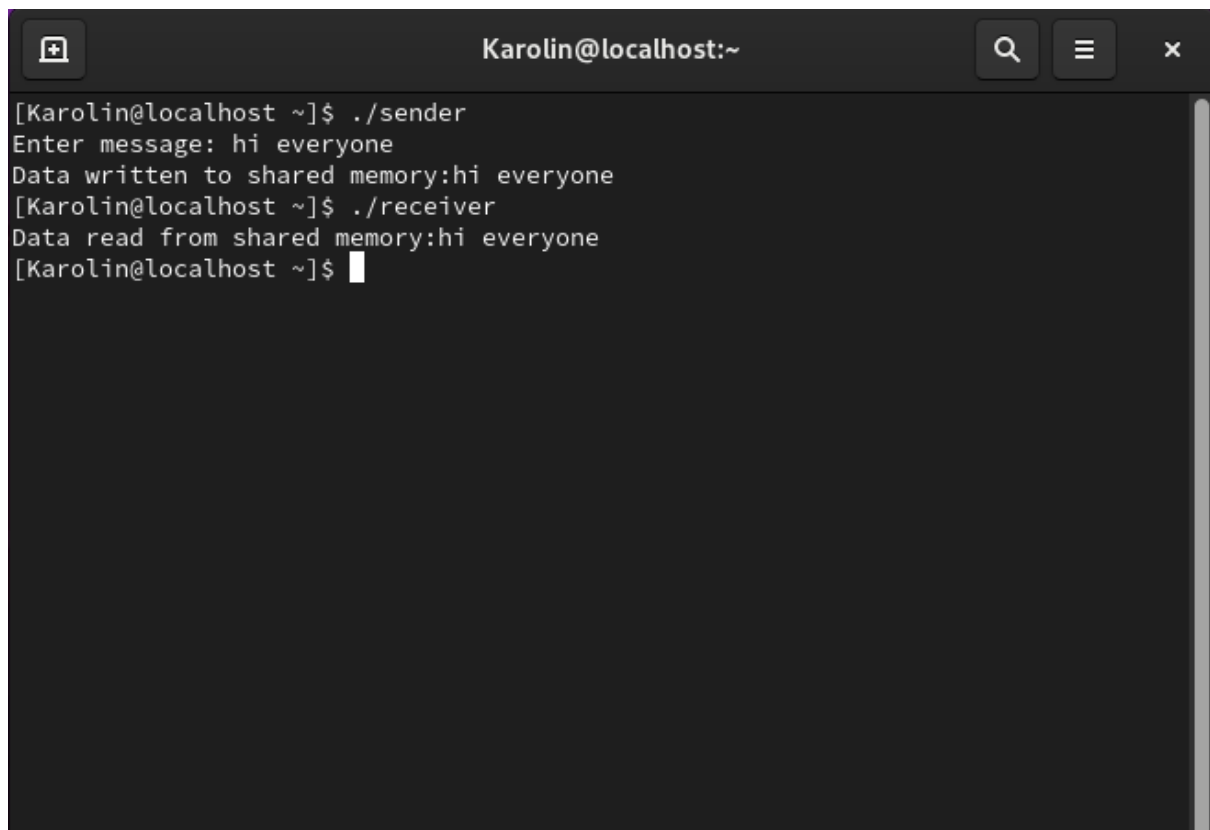
```
Karolin@localhost:~ — vi sender.c
#include<stdio.h>
#include<sys/ipc.h>
#include<sys/shm.h>
#include<string.h>
int main(){
    key_t key =1234;
    int shmid;
    shmid=shmget(key,1024,0666|IPC_CREAT);
    char*str=(char*)shmat(shmid,(void*)0,0);
    printf("Enter message: ");
    scanf("%[^\n]",str);
    printf("Data written to shared memory:%s\n",str);
    return 0;
}

"sender.c" 15L, 316B
```



```
Karolin@localhost:~ — vi receiver.c
#include<stdio.h>
#include<sys/ipc.h>
#include<sys/shm.h>
int main(){
    key_t key=1234;
    int shmid;
    shmid=shmget(key,1024,0666);
    char*str=(char*)shmat(shmid,(void*)0,0);
    printf("Data read from shared memory:%s\n",str);
    shmdt(str);
    return 0;
}

"receiver.c" 12L, 247B
```



A terminal window titled "Karolin@localhost:~" with standard macOS window controls (zoom, search, menu, close). The terminal shows the execution of a sender and receiver program that communicate via shared memory. The sender writes the message "hi everyone" to shared memory, and the receiver reads it back.

```
[Karolin@localhost ~]$ ./sender
Enter message: hi everyone
Data written to shared memory:hi everyone
[Karolin@localhost ~]$ ./receiver
Data read from shared memory:hi everyone
[Karolin@localhost ~]$
```